Newton's Second Law: Introductory examples

1. A car of mass 1200 kg is travelling along a level road. It accelerates (in the direction of travel) at 0.2 ms^{-2}. A constant resistance of 700 N opposes the motion. What is the driving force?
2. The car is approaching a set of traffic lights and decelerates at 1.5 ms^{-2}. The resistance to motion remains 700 N. What is the braking force?
3. The driver brakes harder, increasing the braking force by 250 N. What is the rate of deceleration?
4. A different vehicle slows for the lights, decelerating at the same rate. The braking force is 3100 N and the frictional force opposing the motion is 1400 N. Find the mass of this vehicle.